Project Plans:

Fisher's Little Free Library

By: Drew Fisher

www.FishersShopOnline.com

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Introduction:

These plans are to build the Little Free Library that's depicted in the Fisher's Shop YouTube video that can be seen here: https://youtu.be/uEavnR-oZoY. A friend from church asked if I could repair her old library but it was too far gone. Instead, I chose to design and build her a brand new one! I based the design on a craftsman-style cottage and tried to make it look like a little house. The designs called for a round accent window, which I knew would be a challenge to make! And, in an impromptu decision, I chose to add some character and give it the appearance of having typical exterior siding that you might see on a house. Both turned out fantastic and the library turned out looking amazing! Let me break down each step for you so that you can make one for yourself!

Estimated Cost in required materials: **$275**

Tools Needed:

Basic Tools	Power Tools
NOTE: Orange tool names indicate alternatives	
• Tape measure • Pencil • Marking knife / Razor blade • Rubber mallet • Hammer • Chisels • Straight Edge • Combination square • Speed square • Clamps (squeeze clamps) • Band clamp • Sandpaper • Drill bit set (brad point) • Forstner Bits • 5" Hole Saw • Pull saw • Flush-trim Saw • Caulk gun • Paint brushes	• Table Saw (w/ dado stack or FTG blade) • Miter Saw • Impact Driver • Power Drill • Power Sander • Track Saw • 18ga Brad Nailer • Router (w/ straight and flush-trim bits)

Materials Needed:

- Wood
  - Quantity 1 of 3/4” 4ft by 8ft plywood sheet
  - Quantity 3 of 2x4 premium studs
  - Quantity 1 of 6”x72”x3/4” poplar board
  - Quantity 1 of 4x4 post
- Acrylic sheet 24” x 18” and .093 thickness
- Cabinet door knob
- Neodymium magnets -or- cabinet door latch
- Waterproof Wood glue (ex. Titebond 3)
- 2” & 1 1/2” 18ga brad nails
- 12” piano hinge (USE THIS ONE: https://amzn.to/3YpTVlt)
- Solar-powered LED lights (USE THESE: https://amzn.to/4dlu1n0)
- 2” Round Plastic Louver Soffit Vent: (USE THIS ONE: https://amzn.to/4fq3Nl6)
- Qty. 4 of 1 1/4” wood screws (For base plate)
- Qty. 4 of 3” decking screw (For base supports)
- Qty. 4 of 3” hex lag bolts w/ washer (For mounting)
- Qty. 12 of 1/2” pan-head wood screws (for window & door acrylic)
- Qty. 8 of 1 1/4” wood screws (mounting roof to structure)
- Wood filler
- Silicone sealant
- Exterior caulk (paintable)
- Exterior paint
- Painter’s masking tape
- Flex Seal
- Roofing shingles
- 3/4” Roofing nails
- Roofing shingles

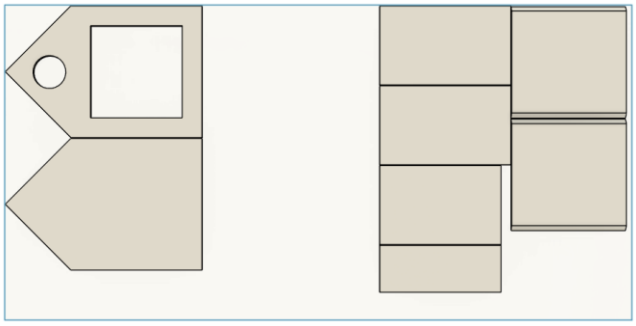
Assumptions:

All the measurements for this project are in Imperial. If you’re one of my metric-loving viewers, I’m assuming you know how to convert our silly American measurements into something you’re comfortable with. Google is a heck of a thing.

I’m assuming you know you can use whatever lumber you have available to you. You’re obviously not required to make this out the same lumber I did. I just chose this option because it was the most cost-effective.

I’m assuming that you have a dado stack for your table saw. If not, I’m assuming that you’ll be using a flat-top grind (FTG) blade and are competent and capable to create the required dados/notches using this blade along with fence adjustments.

Main Structure:

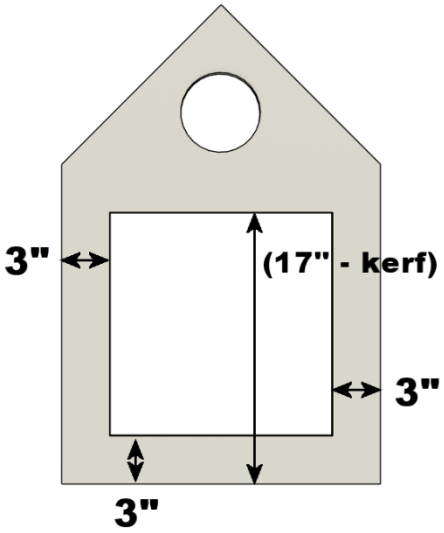


The first thing to know is that all the plywood pieces that you'll need can be taken from a single 4x8 sheet of plywood. If the sheet is too big for you to fit in your vehicle, you can have it cut in half (across the long edge) so you end up with two 4x4 squares.

Use the below table to cut out all the plywood pieces that you'll need.

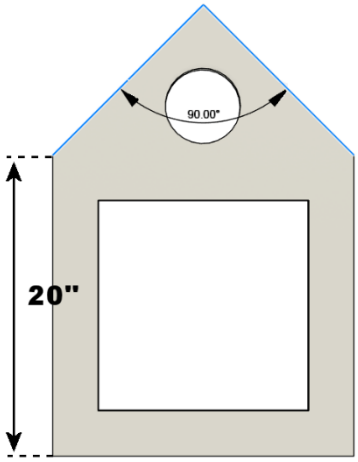
Name:	Quantity:	Dimensions:	Done:
Front & Back	2	30" x 20" x 3/4"	
Sides	2	20" x 12" x 3/4"	
Bottom	1	18 1/2" x 12" x 3/4"	
Shelf	1	18 1/2" x 7"	
Roof	2	17 1/2" x 18" ***see note**	

* - The roof panels will end up being 17 1/2" x 17" once you cut the 45-degree angles on the edges. For now, leave them slightly oversized (18" long) per the cut list.



To cut the opening for the door, I used a reverse plunge on the table saw. **BE CAREFUL** if you use this method as it can lead to kickback if done improperly. The opening is exactly 3" in from either side, 3" up from the bottom, and 17" up from the bottom. If you're using the reverse plunge, then set your fence to 17" **MINUS** the thickness of your blade so that you cut on the inside of your line.

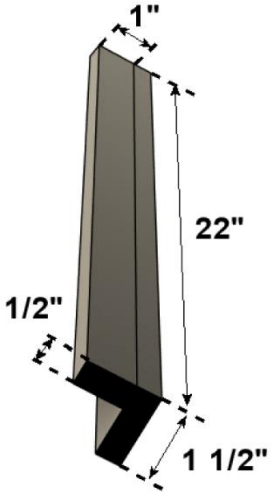
That being said, it's not imperative that the opening be cut perfectly. You could easily cut it out using a jig saw or plunge cuts with a circular saw. Any small imperfections will just get covered with caulk.



The peaks of the front and back need to be identical. The easiest way to achieve this is to temporarily fasten the front and back pieces together with tape. Lay out the two cut lines, starting at 20” up from the bottom on both sides and meeting in the center of the top edge. The angle should be exactly 45-degrees. Make the cut using a track saw or a circular saw with a straight edge.

Next, using some waterproof glue (ex. Titebond3) and an 18ga brad nailer, begin to assemble the main structure by attaching the two sides onto the bottom. After that, lay the assembly down onto its side, add a bead of glue, lay on the front or back piece so that it’s flush all around, and nail it into place. Repeat the same process to attach the remaining front or back piece.

Corner Trim:



For the corner trim pieces, cut out four strips (from the 2x4s) that are 1” x 1/2” x 22” and four strips that are 1 1/2” x 1/2” x 22”. Assemble each pair with glue and clamps using one of each width and orienting them as seen in the illustration to the left.

Once dry, miter one end at a 45-degree angle so that the length of the assembly is exactly 21” at its longest edge. Two should have the angle on the 1 1/2” side going one direction and the other two the other direction.

Make sure each one is clear of glue residue or squeeze-out that would prohibit the pieces from sitting flat against the corner of the structure. Sand the outside edges so that they're smooth and ready for painting in a later step.

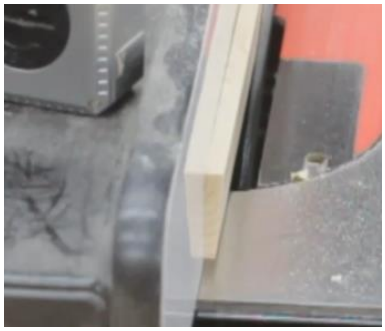
At this point, you can glue and nail them onto the corners of your structure so that the bottoms and top are flush.

Siding:

This step is completely **optional**, but it <u>does</u> add a lot of character to the library. If you wish to go forward with this step, begin by first cutting out all the 1/4" strips from the 2x4s. Keep in mind that the Peak strips vary in length required from a short 2" (at the tip of the peak) to a long 18" (near the bottom of the peak). Only 10 should be needed since the offcuts of each can be used for the other peak.

Name:	Quantity:	Dimensions:	Done:
Side strips	40	1" x 11 1/2" x 1/4"	
Long strips	42	1" x 18" x 1/4"	
Peak strips	10	1" x 20" x 1/4"	

Once all the strips have been cut out, tilt your table saw blade by 5-degrees, adjust your fence to 1/8", and run each strip through again on edge. This will add a small taper to each one.



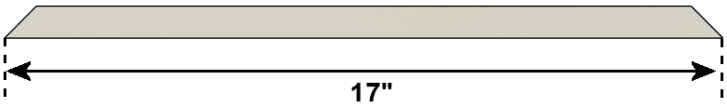
Lay the structure on one of its sides and spread a layer of waterproof glue on the surface between the corner trim pieces. Then, beginning at the bottom, start to lay on the siding strips with the flat side towards the glue and the thicker side towards the bottom.

As you get to the peak, make sure both ends of the strips overlap the angled roof line. Once dry, trim off the excess with a flush-trim saw.

When gluing on the siding strips onto the front, only go halfway up the door opening and let the pieces dry. This way, you can still get your flush trim saw into the opening to trim off the overlap. Then, proceed to glue on the rest all the way up to the tip of the peak. Finally, repeat the same process to apply the strips on the sides of the structure.

Roof:

A 45-degree edge needs to be put onto the edges of the roof panels (**NOT** the 17 1/2” edges). Use your table saw to make these cuts so that the non-mitered edge (tip to tip) is exactly 17” long.



Once the roof panels are ready, cut out the pieces for the fascia that will surround the roof structure.

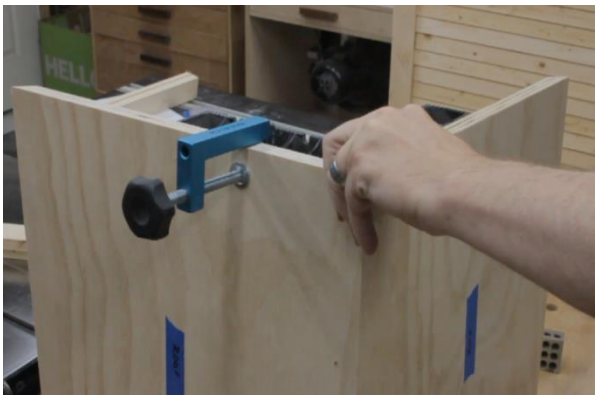
Name:	Quantity:	Dimensions:	Done:
Peak Fascia	4	17” x 2” x 3/4” (*)	
Side Fascia	2	17 1/2” x 2” x 3/4”	
Block Fascia	4	2 13/16” x 2” x 3/4”	

* - The Peak fascia boards have a 45-degree cut on both ends, and the 17” dimensions are from tip to tip.

Glue and nail the two roof panels together so that they're perfectly 90-degrees and square with one another.

Next, attach the pairs of peak fasciae to one another using glue and nails.

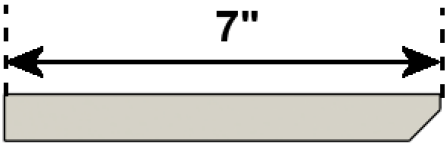
Once the roof panels have dried, run a bead of glue along the front and back edges of the assembly and nail on the peak fascia so that it's flush along all the top edges.



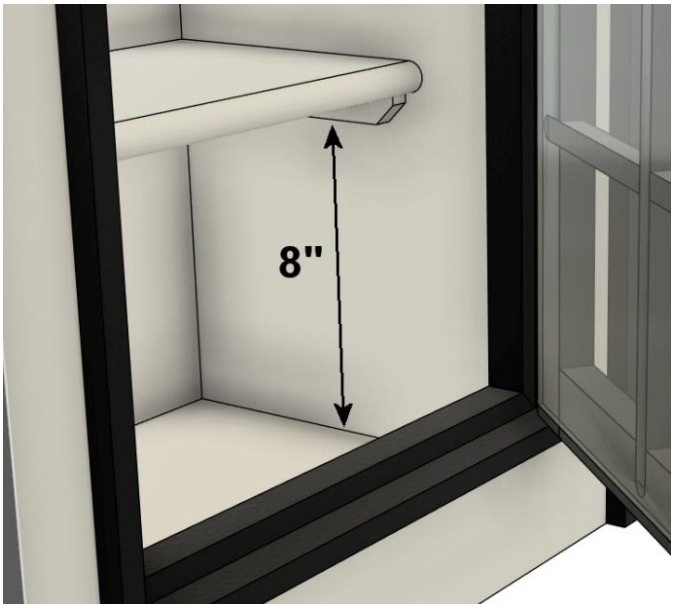
Attach the block fascia pieces to the ends of the side fascia with glue and nails. Then, glue and nail on that assembly onto the sides of the roof so that it's flush with the edge of the roof line as well as the front and back peak fascia pieces.

Shelf:

The shelf within the library needs two cleats to sit on. Cut two strips of wood from your 2x4s that are 3/4” x 1/2” x 7”.



It’s up to you, but I put a small chamfer on the bottom front corner and sanded those edges, so they won’t scrape any fingers.



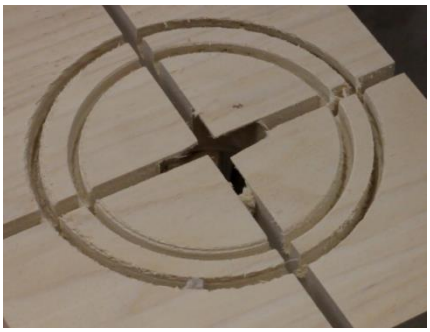
Each cleat gets mounted 8” up from the bottom of the library. Attach with glue and brads.

For the shelf itself, I chose to edge-band the front with a thin strip of 2x4 so that it would paint better in a later step. However, you could add a roundover (seen in the illustration) or just leave it as a square plywood edge.

In the end, I chose to not fasten the shelf down onto the cleats in case the owners wanted to remove it for larger books. I don’t think they ever will, but I left it as an option.

Accent Window:

This step is completely optional, but similar to the siding, it adds a lot of character and looks great! You can make this with ordinary power tools, but it’s not easy… especially if you have a sub-par hole-saw kit like I do. If you have a CNC, then I recommend just using that. If you don’t want to risk it, then I suggest just making it a square or rectangular accent window to achieve a similar look. Regardless, make it from a **harder wood**, like poplar so that it holds up better during the construction process.

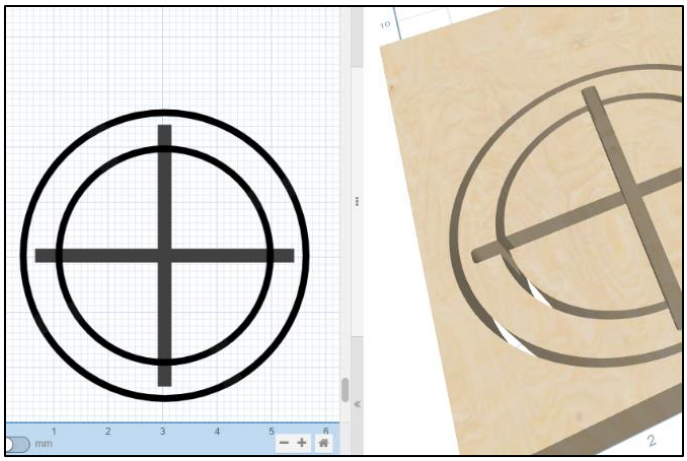


To make it **without a CNC**, take a square piece of material that is 6” x 6” x 1” and cut two dados into the bottom of it that are exactly 1/4” wide and 3/4” deep. This forms a cross on the bottom of the piece. Next, mark the exact center of where the two dados intersect and use a 5” hole saw to cut a circle out of the piece, then again with a 4” hole saw. Cut along these hole saw kerfs (if you didn’t cut all the way through) with a bandsaw or jigsaw and use a combination of the belt sander and spindle sander to touch it up.

Remember, the edges don't have to be perfect since everything gets caulked in a later step.

Then, to make the hole in the front of the library, you'll have to trace the window casing piece and cut it out with a jigsaw. This is because the traced circle will be less than 5" since you have to subtract the kerf size of the hole saw. Make the hole so that the center is **6 3/4"** down from the peak of the front.

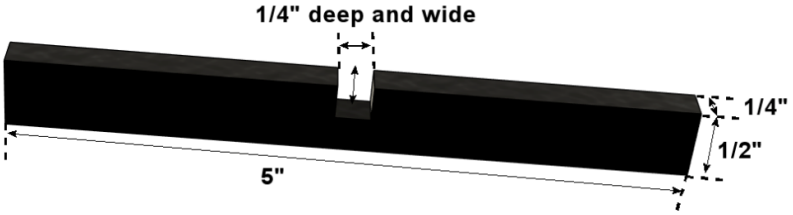
To make it **<u>with a CNC</u>**, design something similar to the following using your CNC software and have the machine cut it out.

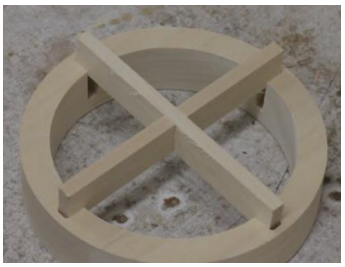


Cut on the outside of the larger 5" circle, on the inside of the smaller 4" circle, and make the dados 3/4" deep by 1/4" wide. I used a 1/8" end mill to make the cut.

The nice thing about using a CNC, is that you can use the 5" hole saw to make the opening for the casing on the front of the library. No need to trace it and use a jigsaw. Make the hole so that the center is **6 3/4"** down from the peak of the front.

Regardless of how you made the window casing, the dividers come next. To make the small dividers, first start by cutting a 1/4" wide and 1/4" deep dado in a piece of stock that is 5" long and 1/2" thick. Next, cut two 1/2" strips across your dado to form each individual divider. These pieces can then be put together into a cross. Trim the ends so they can fit into your window casing, glue, and tap them into place.



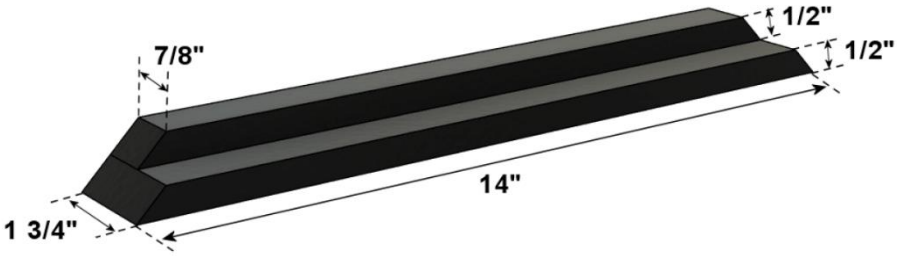


Next, for the acrylic back, I suggest simply tracing your window casing onto the acrylic and rough-cutting out the circle using a bandsaw or jigsaw. Refine the shape on the belt sander and make it slightly smaller than the casing itself so that it won't catch and get caught on the hole when mounting the window casing later. Then, drill several mounting holes and use small screws to attach it to the back of the window casing.

Vent:

I'm not certain if it's needed or not, but I know it can't hurt. I purchased a small 3D-printed vent off Amazon (link in the materials list). I drilled a 2" hole using a forstner bit that was 4" down from the peak of the back of the library. Once the hole was drilled, the vent cover is just friction-fit and pressed into the hole. It can be easily removed for painting in a later step.

Door Casing:

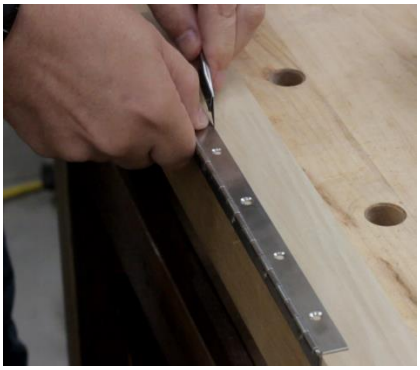


Since the door and the door casing will see a lot of use, I recommend you use a **harder wood**, like poplar, to construct it.

The door casing is made up of eight pieces, four of which should be cut to 1 3/4" x 1/2" x 15" and four that are 7/8" x 1/2" x 15".

Name:	Quantity:	Dimensions:	Done:
Outer Casing	4	1 3/4" x 1/2" x 15"	
Inner Casing	4	7/8" x 1/2" x 15"	

Next, you need to cut the inlay for the piano hinge into one of the sides of the outer casing. Take an outer casing piece and lay it down flat with the piano hinge straddling one of the edges. Center it on the piece and then hold it firmly while tracing the outside of the hinge with a marking knife or razor blade.



Using a router with a straight bit, set the depth of cut to the thickness of the piano hinge, and remove the bulk of the material up to your knife line.

Finally, with a mallet and chisel, chop down to accentuate your knife line and then chisel out the remaining bit of material to finish out the inlay. Test fit the piano hinge into place and verify that it fits snugly and is flush to the face of the board. Don't mount the hinge in place just yet.

Once you have the pieces cut out, glue them together so that the 7/8” wide face of the inner casing is against the 1 3/4” face of the Outer casing and flush to the back edge. When these pieces have dried, miter the ends at 45-degrees so that the entire piece is 14” long from tip to tip.

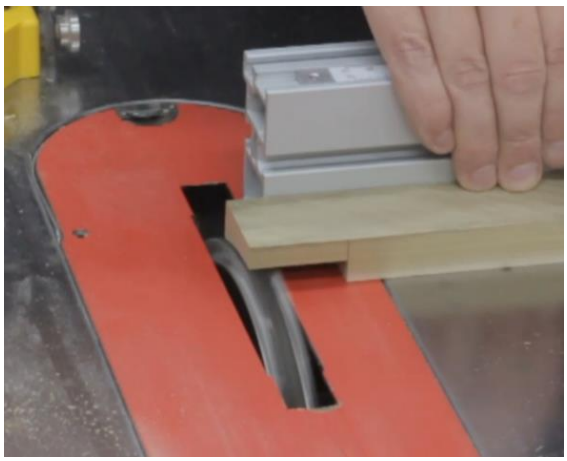
I recommend using referential measurements so that you can sneak up on the perfect fit. Once you have all four pieces of casing dry-fit and looking good within the opening of the library, take them out and glue them up using a band clamp. When that dries, glue the casing into the door opening so that it is coplanar with the corner trim. Then, secure it further with a couple brads into each side.

Door:

Similarly to the accent window and the door casing, the door should be assembled using a **harder wood** like poplar so that it holds up to usage better over time.

Begin by first cutting out the required pieces to form the frame of the door. You’ll need four pieces that are 12 3/4” x 1 1/2” x 3/4”. Additionally, you’ll need two pieces for the dividers which will be 11 1/4” x 1/2” x 1/2”.

Name:	Quantity:	Dimensions:	Done:
Door Frame	4	12 3/4” x 1 1/2” x 3/4”	
Door Dividers	2	11 1/4” x 1/2” x 1/2”	



Next, using a dado stack or an FTG blade in your table saw, create the half-laps on either side of each door frame piece. Make sure that the depth of cut is exactly 3/8” deep and that each half-lap is 1 1/2” wide.

Then, using waterproof glue, assemble and clamp the door pieces together. Make sure all the joints are tight and that the door is square before leaving it to dry.

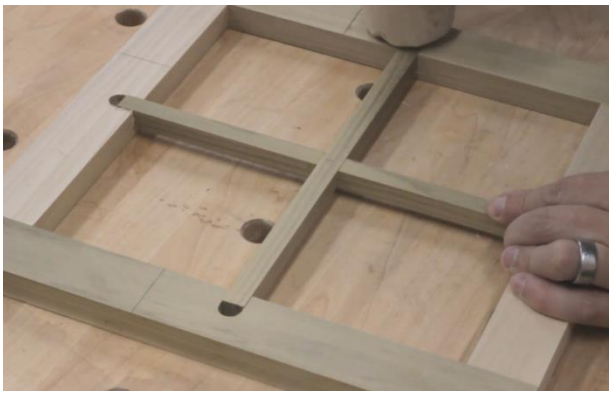
The next step is to mark where the center of each inner edge is on the door frame and to clamp on a temporary fence so that your router can cut in a groove for the dividers at each of those locations.

Use a 1/2” straight bit and begin at a shallow depth while making 3/4” long grooves on both inside edges of the frame. Gradually increase your depth of cut until eventually you’re at 1/2” deep.

Reclamp the temporary fence going in the other direction and repeat the same process to cut the other two remaining grooves for the dividers.

Similar to the dividers you made for the accent window, cut a 1/2” wide dado that is 1/4” deep in the center of each one. This allows them to be assembled into a cross.

Next, add a drop of glue into each notch you cut in the door frame and tap the dividers into place.

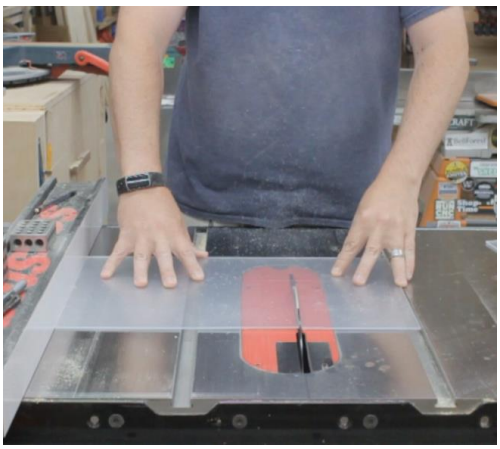


Just like you did with the door casing, you need to inlay the piano hinge onto the edge of the door. First, place the door into your door casing and shim it up so that it’s perfectly centered vertically. Then, mark where the top and bottom of your existing hinge inlay meets the door. At this point, use the same technique as you did earlier to inlay the hinge. Mark around it using a knife, route out the majority of the waste, and chisel out the remaining material.

Drill pilot holes and screw the hinge onto the door. You’ll have to remove it later during the painting step, but you need to mount it now to find where the mounting holes need to be drilled on the casing.

Be aware that the mounting screws for the hinge are too long to go into the casing without poking through the other side. I shortened the screws that go into the casing by about 1/8” by just sanding off the tips on the belt sander.

Next, hold the door up and line up the other half of the hinge within the inlay of the casing. Pre-drill the mounting locations being careful not to poke through the other side. Once you have these locations drilled, you can proceed on to the next step prior to mounting the door.

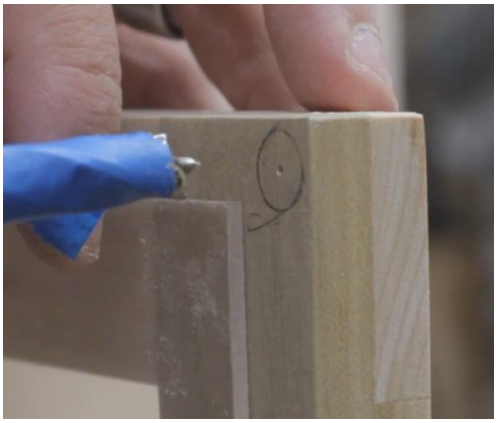


Cut the piece of acrylic for the back of the door. The dimensions of the piece should be 12” x 12”, however you may need to adjust those measurements depending on what you choose for a doorknob.

Next, center the piece of acrylic on the back of your door, drill eight mounting holes, and fasten the piece onto your door with small wood screws.

With the acrylic put on, you can finish mounting the hinge to the casing.

Then, you can mark, drill, and mount whatever kind of doorknob you’ll be using for the library. Be advised that if you center it, you may be drilling into the notch on the back with the divider or into the acrylic. Position your doorknob so that neither of these will be an issue and mount it in place.

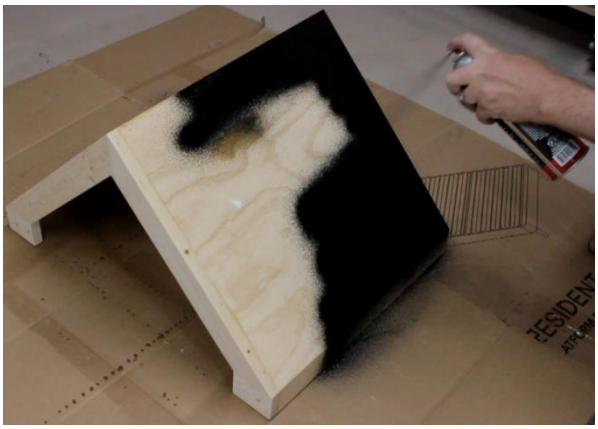


En lieu of a latch, I chose to use neodymium magnets to hold the door closed. They work very well and they’re low profile. If you decide to go this route, consider that you only have 1/8” of room to work with. If two of your magnets have a combined thickness that exceeds 1/8”, you’ll have to inlay them. I drilled shallow holes using a brad point bit and superglued and pressed them into place. For the door casing magnets, I simply placed them onto the door magnets, added a drop of superglue, and then closed the door. After about a minute, the glue had cured and held the magnet in place on the casing. This assures that they’re perfectly lined up.

Caulking:

Try your best, and caulk the rest! I used exterior-grade paintable window and door caulking to completely seal up the outside. Every seam, every crack, every knot hole got completely sealed. Take your time to make sure you get every gap filled and then smooth it out with a caulking tool or your finger.

Roof Sealing:



Instead of having to purchase an entire roll of roofing felt or tar paper for this small project, I picked up a can of Flex Seal. This is essentially liquid rubber that you can spray onto a surface to make it completely waterproof. Make sure you're in a well-ventilated area (preferably outside) and spray on a heavy coat onto the entire roof. Don't spray the fascia since we'll be painting that in the next step.

Painting:

Take a moment and fill all the nail holes that will be visible with a wood filler. Wait until dry and sand smooth. For a paint, I made sure to pick out a very hardy exterior paint and I brushed on two coats on everything. I started with the white, which was the color I had picked for the siding and the interior. Don't worry too much about getting your siding color on the trim, the window casing, or the door since you'll be taping and painting those immediately following this part. Just be sure to get things fully covered so that the siding has the best protection against the elements that it can get.

Once the siding and the interior are completely dry, start taping off the trim and all the other areas that you don't want to get paint on. Use a scraper to wedge the tape up into each ridge of the siding as you tape off the corner trim. Use lots of little pieces to go around the curves of the accent window. Remove the acrylic from the window and the door. Remove the hinge and tape off the inlay portions. Remove the vent. Lastly, tape off or remove the doorknob.

You shouldn't have to worry about taping off the roof fascia since it won't matter if you get any on the top. I chose to go ahead and paint the underside of the roof the same color as the trim, but even that shouldn't matter because it won't be seen. Once you have it masked, put on two coats and let things thoroughly dry.

When the paint is dry, you can pull off the masking. If you're lucky, you'll have super clean lines. I still had a handful of areas which needed some touch-ups, and I did that with an artist's brush.

Silicone:

To prevent moisture and water from getting in through the window or the door, I made sure to seal them with a bead of silicone on the **inside**. For the accent window, I ran a bead around the outer edge of the circular acrylic piece, dropped the piece back on it, lightly pressed down, and then gently put the screws back in without overtightening and squeezing all the silicone out.

I did the same thing with the inside of the door. I placed a single uninterrupted bead of silicone just inside the outer perimeter of the acrylic sheet, dropped the acrylic piece back down, and re-secured it with the mounting screws, making sure not to overtighten and squeeze the silicone too much.

Lights:

I thought it would look neat if the library lit itself up at night, so I purchased a set of solar-powered LED lights (link in the materials list). These instructions are for that particular set but I'm sure you could adjust them as needed to make a similar set work. The spike on the solar panel mounting post can be removed and discarded. Next, drill a centered 5/8" hole into the back of the library at about 10" down from the top peak. Then, since all the lights are connected in one long strand, fish the entire line of lights through the hole. Once you get them all pushed through, you'll be left with the larger connecting wire and the post. The 5/8" hole is big enough for both to fit snugly. Add some CA glue to the lower portion of the post and twist it in and hold it so that the solar-panel mount is square to the wall. Once it dries, seal the gaps with caulk and smooth it out with your finger. Next, I used a few command strip hooks on the inside to wind the strand of lights around so that they were held securely in the upper section of the library.

Shingles:

It's time to finally mount the roof assembly! Place the roof onto the library and mark a line that is 3 1/8" in from both the front and rear edges. With the width of the fascia, the 2" overhang, and half the thickness of 3/4" plywood, this should make your mark so that it's centered on the edges of the front and back walls. Drill 4 mounting holes on each roof panel and secure the roof to the library with 1 1/4" wood screws. It shouldn't be necessary to glue or silicone down the roof, but you certainly can if you think it'll help. I secured mine with only screws just in case I ever needed to remove it to work on the library in the future. Next, put a dot of silicone on the top of the screw heads to seal them up before moving on to put down shingles.

For shingles, I chose regular roofing shingles. When I was at the big-box store, there were a lot of packages that were torn or had some missing. The store can't sell these at full price, so you could probably save some money if you bought one of the damaged or depleted packages because all you need is about nine of them.

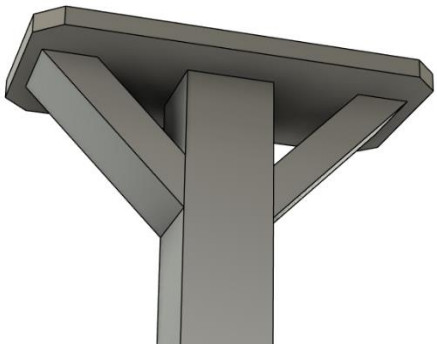
The first shingle should be a solid one, without the grooves cut into it. This is because the very next one will have those grooves and water could easily get down underneath if you don't have a solid one there to protect the roof. Using a razor and a straight edge, cut a shingle that's 19 1/2" long. You just need to score a line on the back and bend the shingle for it to come apart on your cut. Position it on the roof so that there's a 1/4" overhang on the front, back, and side and then secure it in place along the top edge with three 3/4" roofing nails.

The next three shingles also get cut at 19 1/2" and are laid out in a staggered pattern so their grooves don't line up. Make sure to overlap them (as seen in the illustration) to allow for proper water runoff. The last one will have a section that needs to be cut off at the ridge line.

Once you have one side done, repeat the same thing for the other roof panel.

For the ridge line, cut out five smaller shingles that have no grooves in them. Make each one approximately 7"-8" square. Start at each end, fold them over the ridge, and use two 3/4" roofing nails to secure them in place on the inside edge of the shingle. Overlap the nails with the next shingle and proceed to mount them working towards the center. For the center shingle, secure it with four nails. Lastly, use the Flex Seal to rubber coat the heads of the four exposed roofing nails of the center shingle.

Base:



There's a number of different ways you can mount this library, but I'll share how I did it onto a standard 4x4 post. First off, if you'll be putting it on a post, make sure you concrete the post in so it's completely stable and can easily hold the weight of the library without wobbling around.

Cut out a plywood mounting plate that is 8" x 16" x 3/4". You can chamfer the corners if you wish, but they will rarely be seen. Using 2x4s, cut out two small sections that are 8" long and have 45-degree miters on both ends. Secure one end of each support to the plywood mounting plate with glue and a couple long 3" decking screws screwing down through the plate and into the support at an angle. Make sure the screw heads are flush or below the top surface of the mounting plate. Next, straddle the assembly onto the 4x4 post, verify that it's level, and drive in 3" lag bolts with washers through the supports into the 4x4 post. For additional support, you can drive a couple screws down through the center of the mounting plate into the end of the 4x4 post.

At this point, you can place the library on top of the mounting plate and use four 1 1/4" wood screws in each corner of the mounting plate to screw up through it and into the bottom of the library.

That's it! Congratulations! Enjoy your new library!

Thanks:

I sincerely hope you enjoyed the video, the build, and the outcome! Thank you so much for supporting Fisher's Shop by purchasing these plans. I appreciate it very much and I'm grateful. I value each one of my viewers and especially those that go the extra mile to purchase the plans and replicate something that I designed and built. My channel grows primarily by word of mouth, so if you've enjoyed my videos and had fun building this project, please consider helping the channel grow by spreading Fisher's Shop videos to your friends and followers on social media. Thanks again!!

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